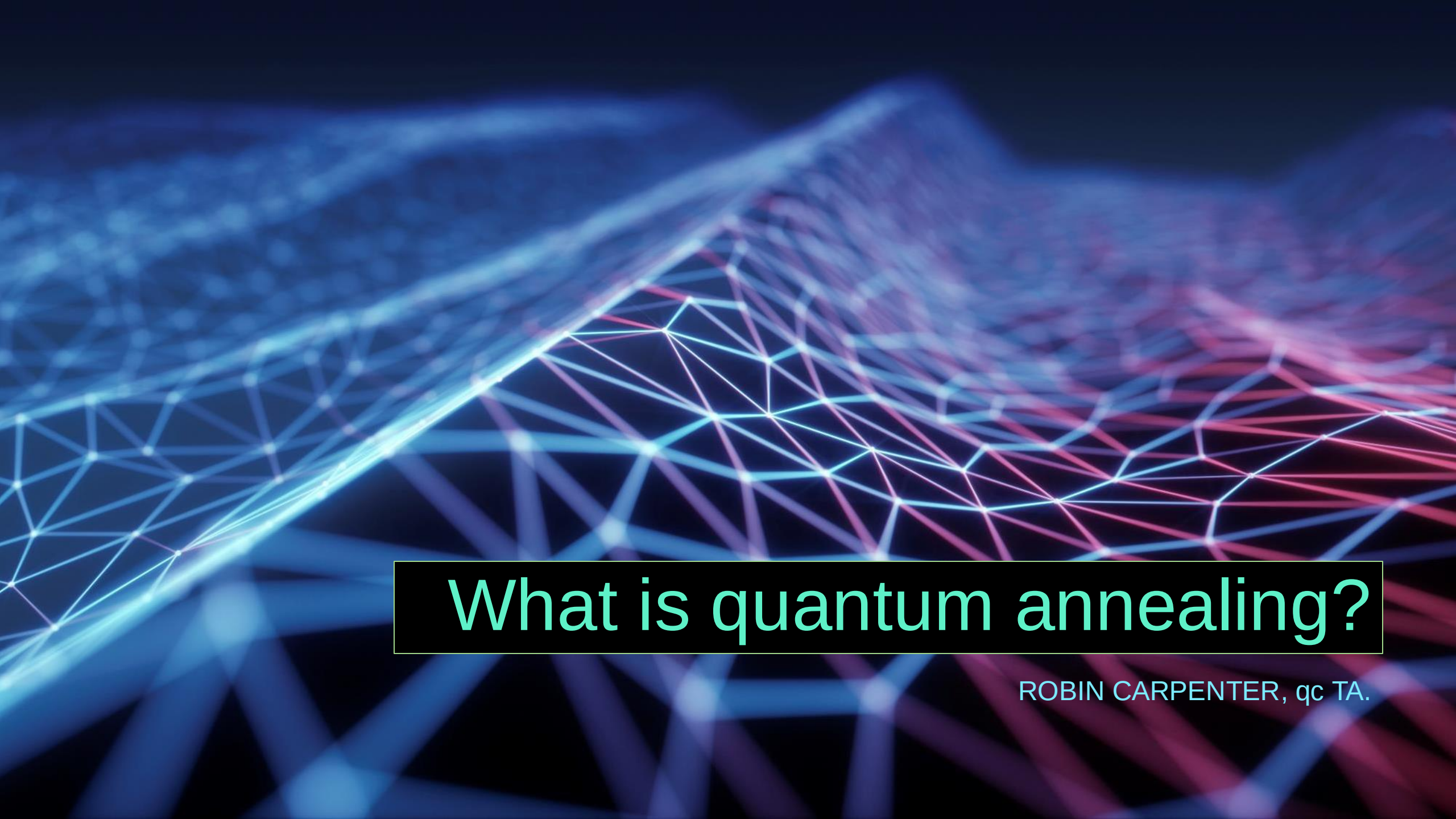


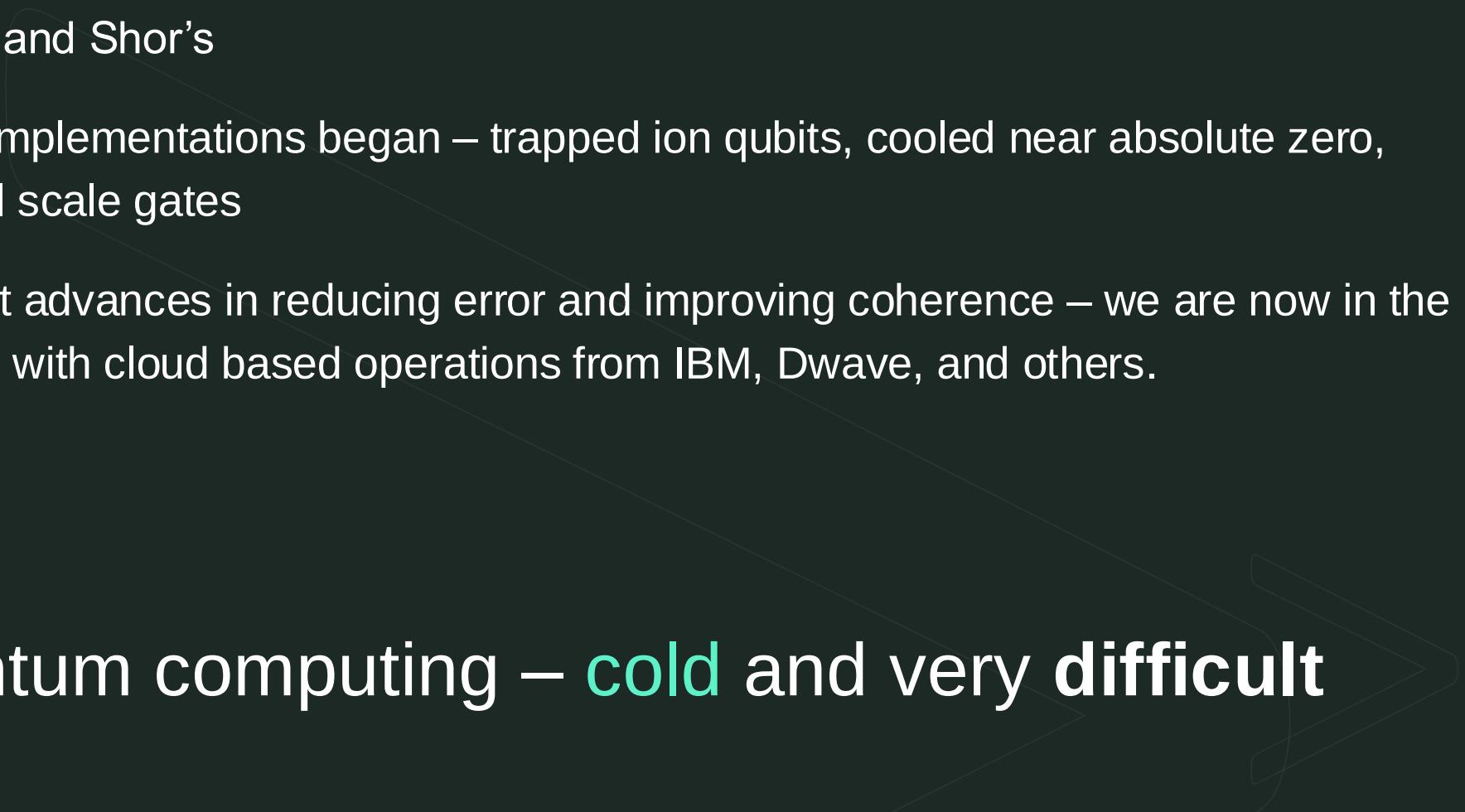


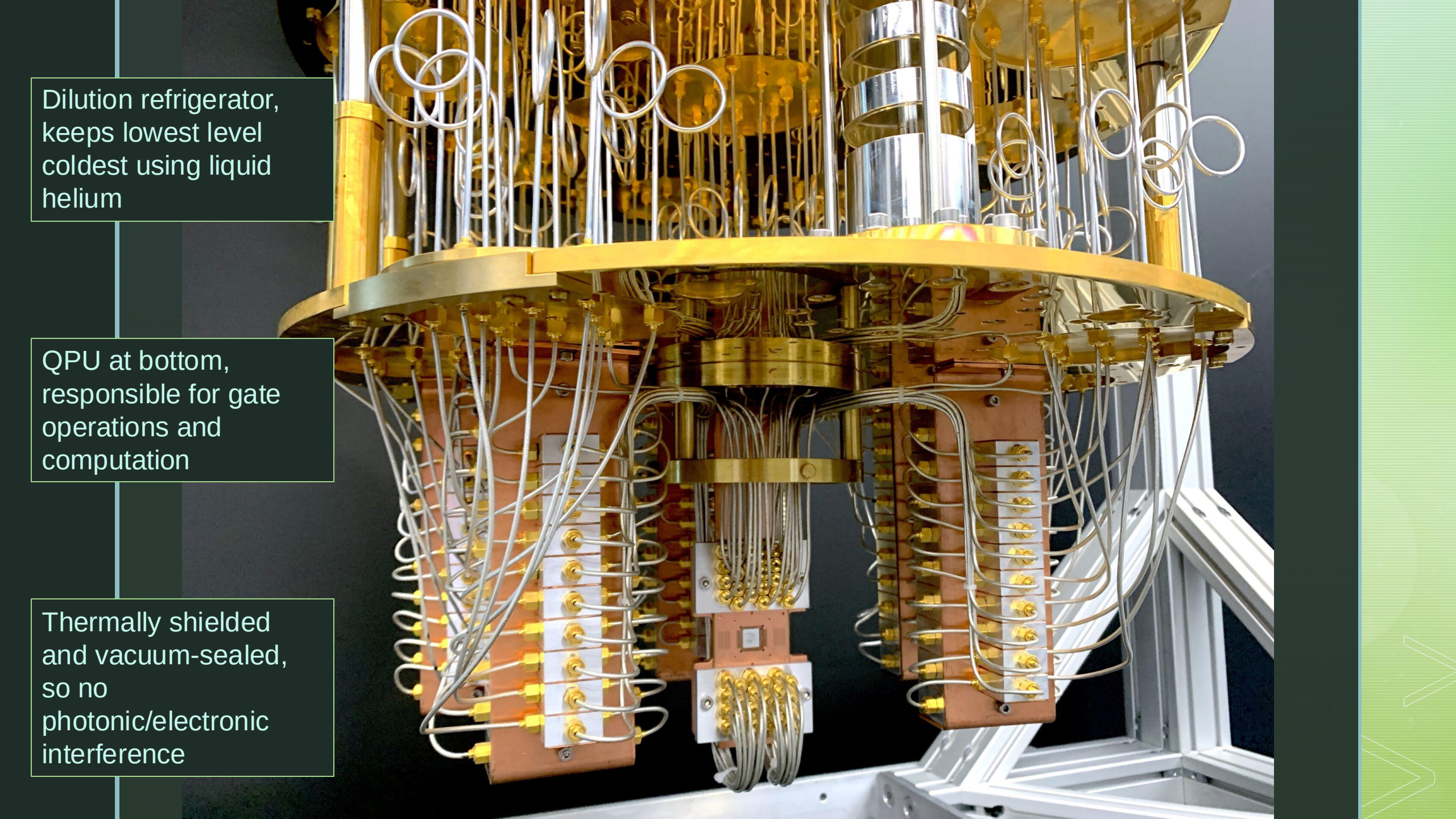
What is quantum annealing?

ROBIN CARPENTER, qc TA.

- 
- Quantum Turing machine first proposed by Paul Benioff, 1980
 - Important quantum algorithms proposed around 1980-1990s, such as Grover's, Deutsch's, and Shor's
 - Physical implementations began – trapped ion qubits, cooled near absolute zero, very small scale gates
 - Significant advances in reducing error and improving coherence – we are now in the NISQ age with cloud based operations from IBM, Dwave, and others.

Quantum computing – cold and very difficult



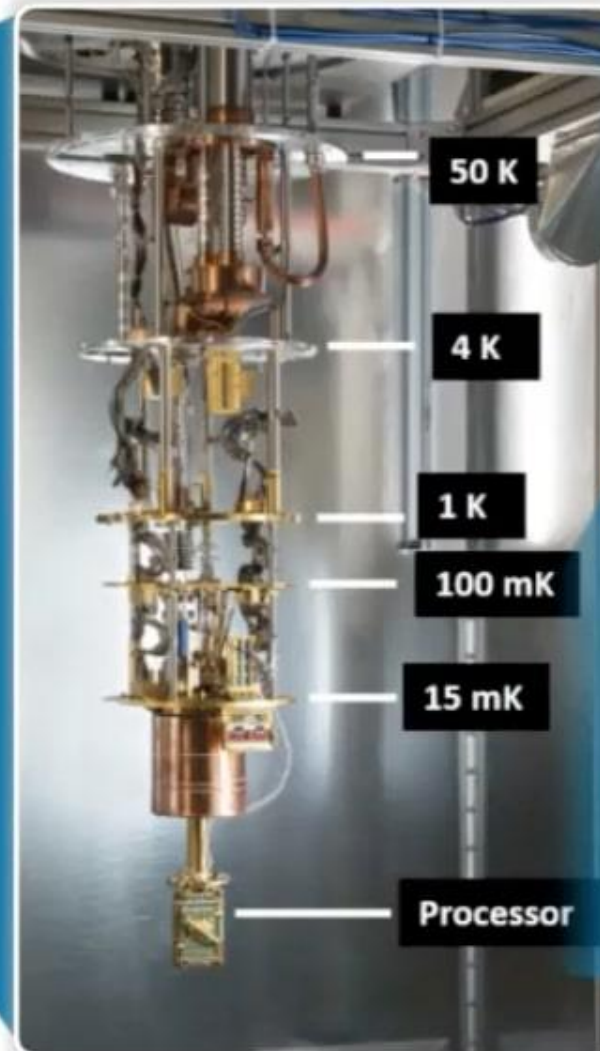


Dilution refrigerator,
keeps lowest level
coldest using liquid
helium

QPU at bottom,
responsible for gate
operations and
computation

Thermally shielded
and vacuum-sealed,
so no
photonic/electronic
interference

What Is A Quantum Computer



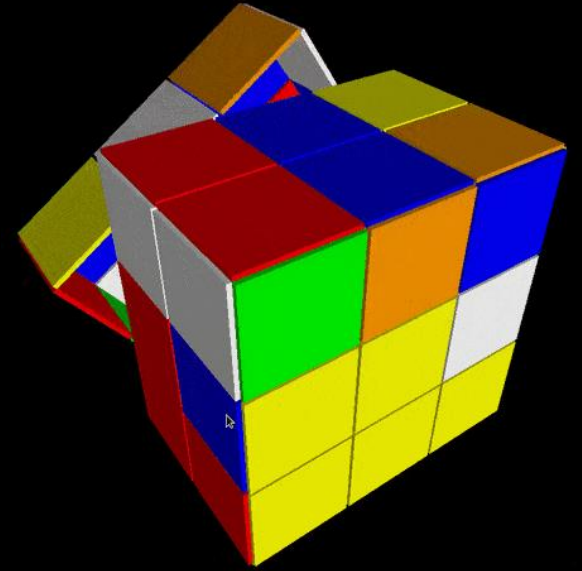
What makes annealing unique?

Annealing uses optimization foundationally

QA is very well suited for combinatorics, where **every possible** solution of a problem is probed to find the best.

By doing the computation slow enough, quantum tunneling and the adiabatic theorem will ensure that the best solution is found.

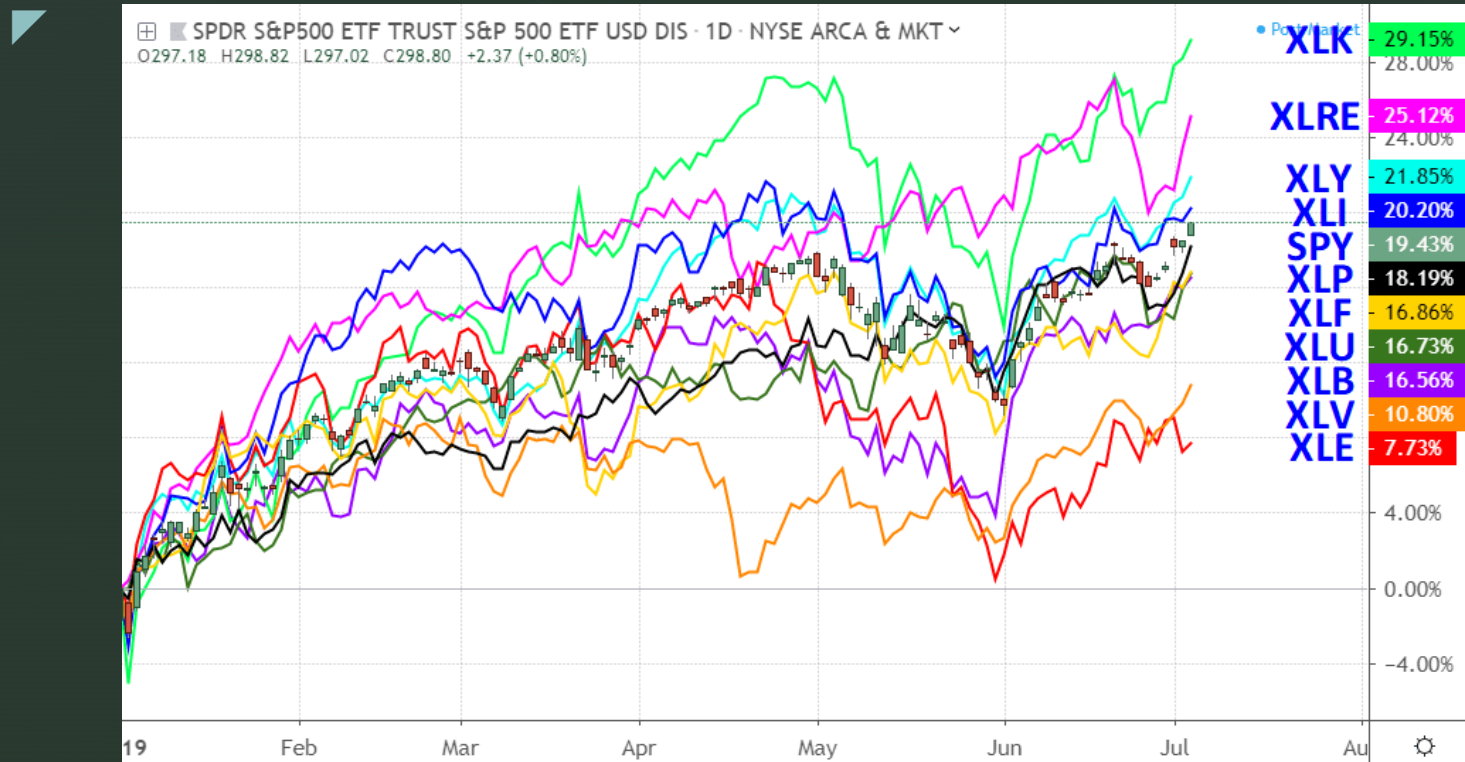
Keep in mind – this exact process can be simulated on computers already!



What are its best uses?

Anything involving rugged “energy
landscapes.”

Finance / Molecular simulation / logistics

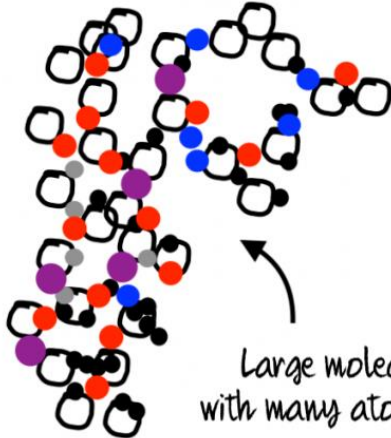


What computers can simulate today



Small molecule with a few atoms

What quantum computers could simulate

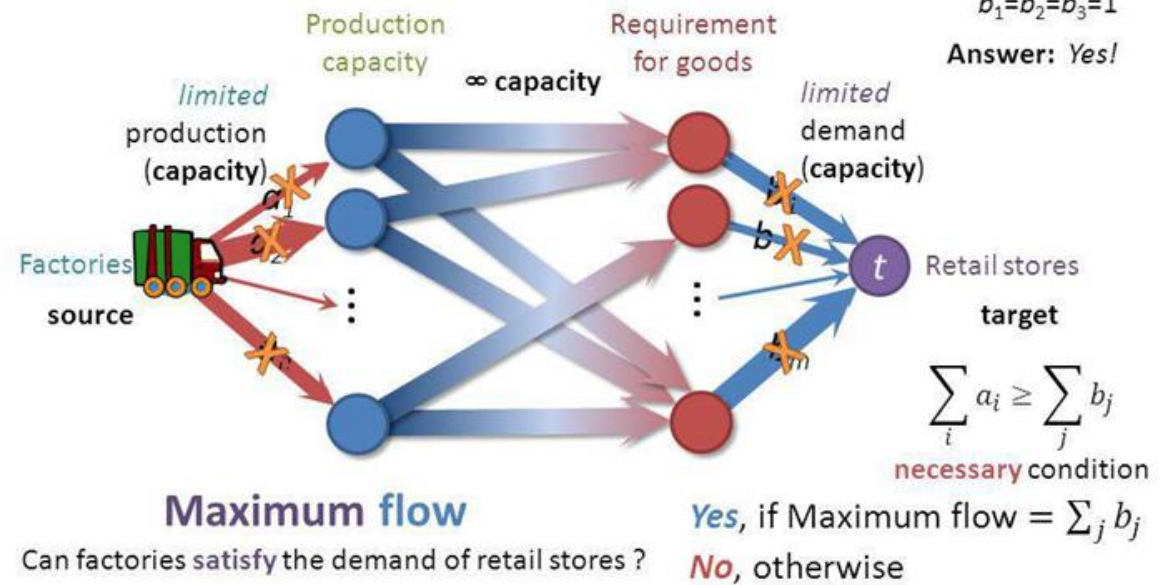


Large molecule with many atoms.
I don't know, maybe a protein or something

thinkingofutils.com

*Considering all atoms and electrons with quantum mechanical effects

• Transportation problem



OK, but how?

THE ISING SPIN EQUATION


$$H = - \sum_{i,j} J_{ij} \sigma_i \sigma_j + h \sum_i \sigma_i$$

What do we see here?

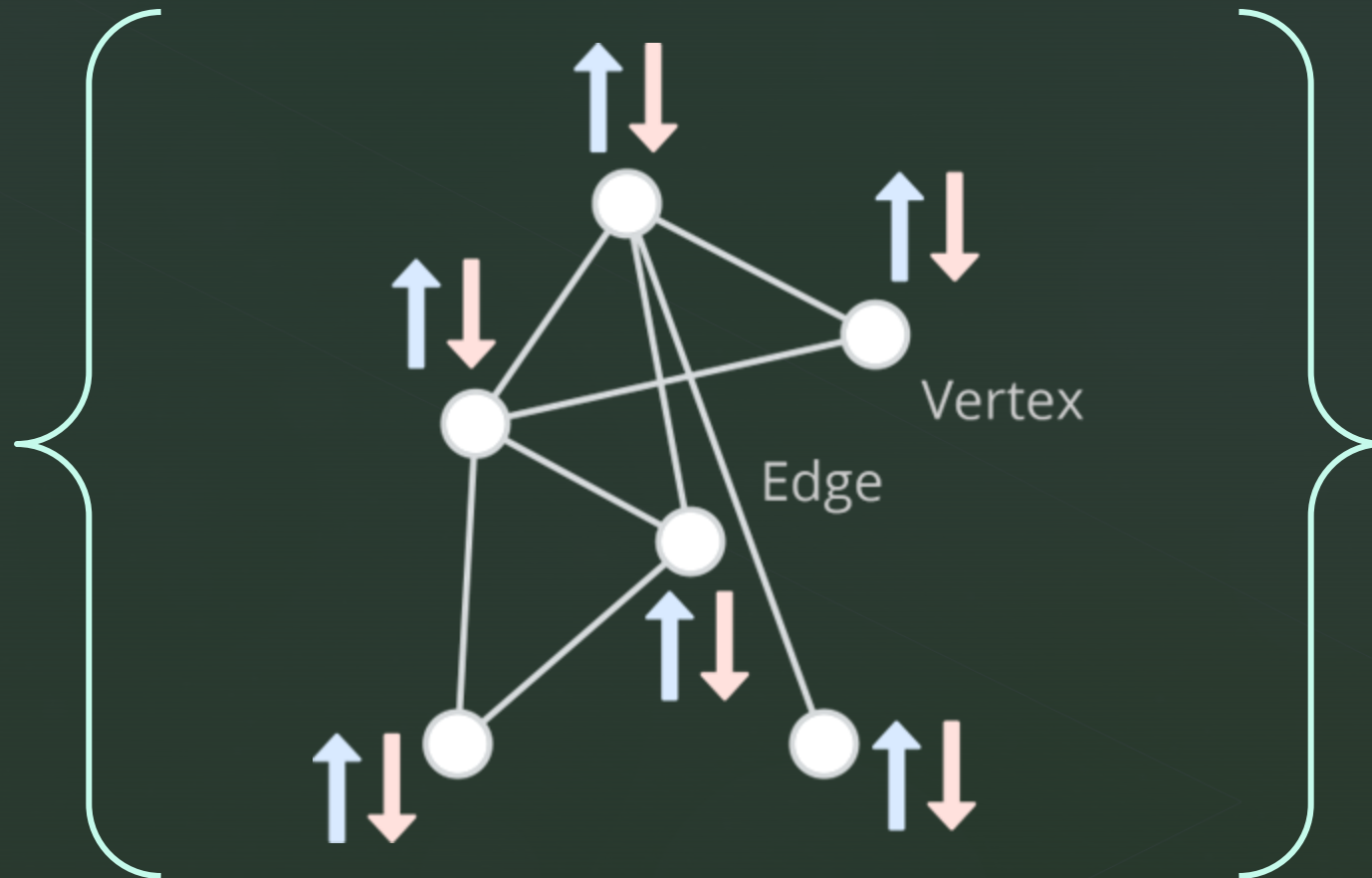
Why two summations?

Why i and j? What do they mean?

Summation 1:
considers all of
the couplings
between each
qubit (edges)

$$H = - \sum_{i,j}^1 J_{ij} \sigma_i \sigma_j + h \sum_i^2 \sigma_i$$

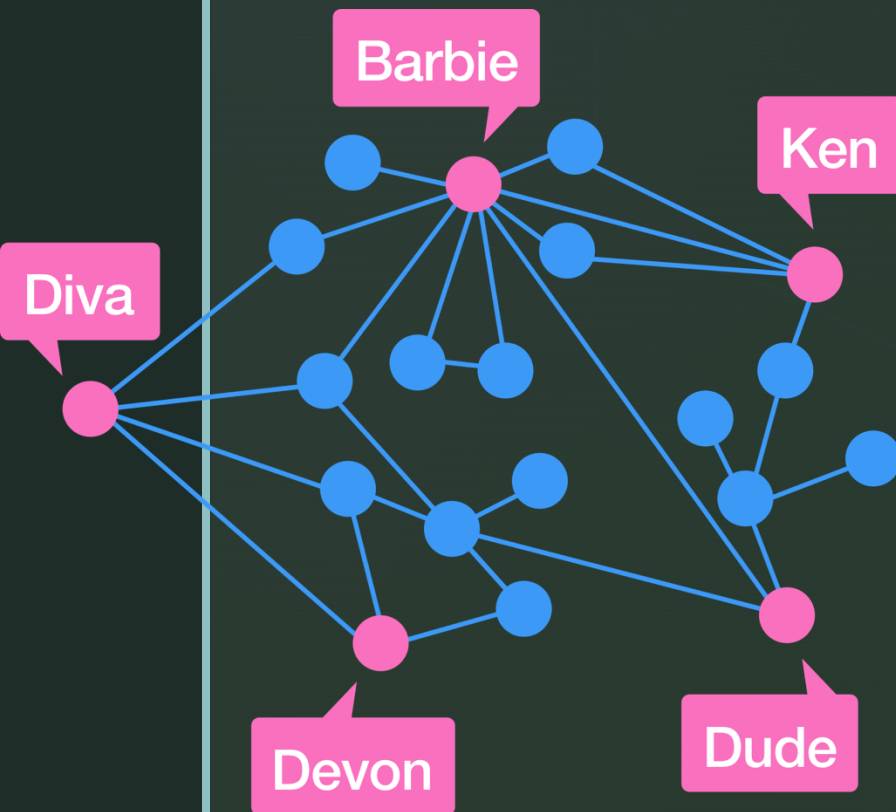
Summation 2:
considers each
individual bias of
every qubit
(graph vertexes)



That's it – we encode the entire problem into the Ising equation!

- Quite powerful – any problem can be broken down into a simple network. Just unknown elements interacting with each other knowingly.
- Keep in mind, a perfect graph has full connectivity – each qubit links to every other.
 - $4 \mid 3+2+1 = 6$
 - $6 \mid 5+4+3+2+1 = 15$
 - $10 \mid 9+8+7+6+5+4+3+2 = 45$
- As n increases, number of connections m approaches $\frac{1}{2} * n^2$

Example: social network

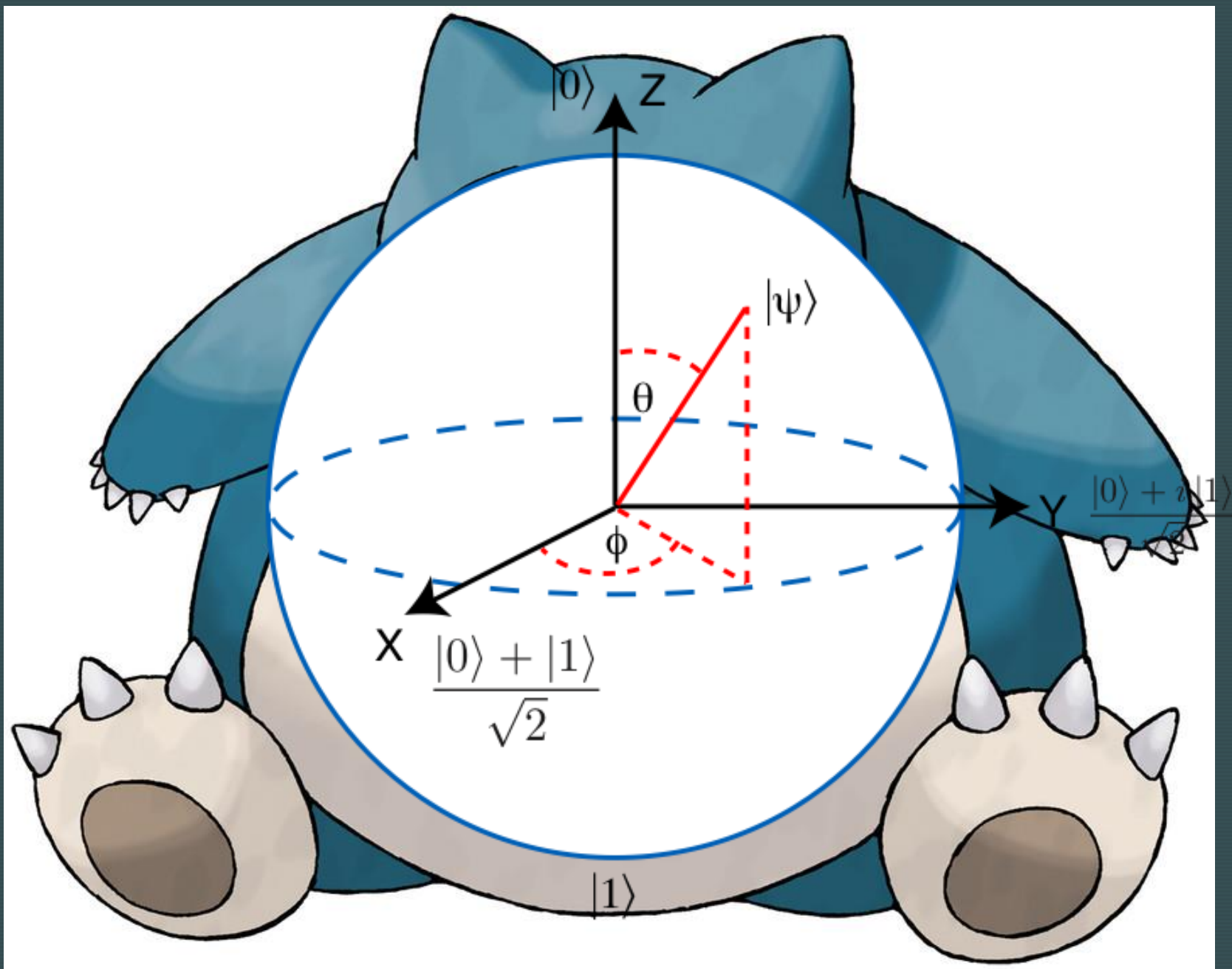


1. Define problem. I want to choose friends (1s) and enemies (0s) from a random group of people.
2. I. Biases: how much I agree with someone.
3. Couplings: how much people agree with each other.
4. Now do it!

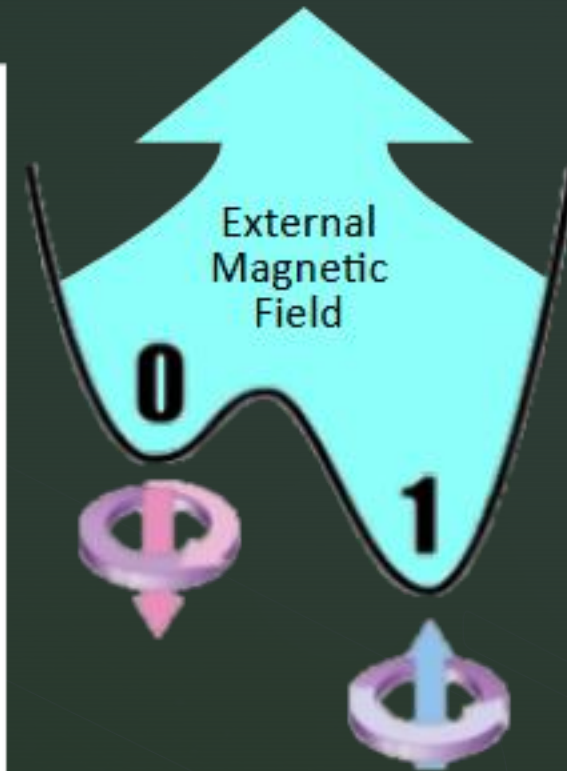
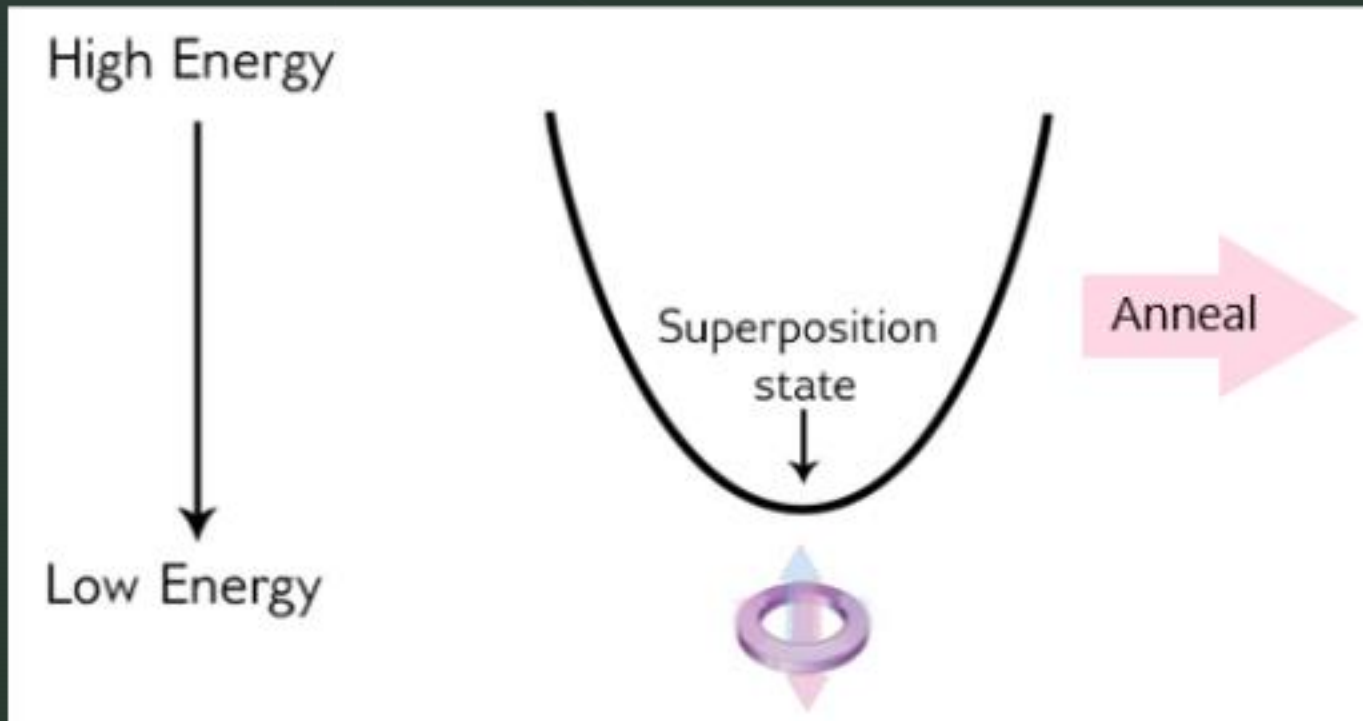
$$N=5, H = -([\sigma_0 - 2\sigma_1 + 2\sigma_2 - 4\sigma_3 + \sigma_5] + [2\sigma_0\sigma_1 - 10\sigma_0\sigma_2 + \dots])$$

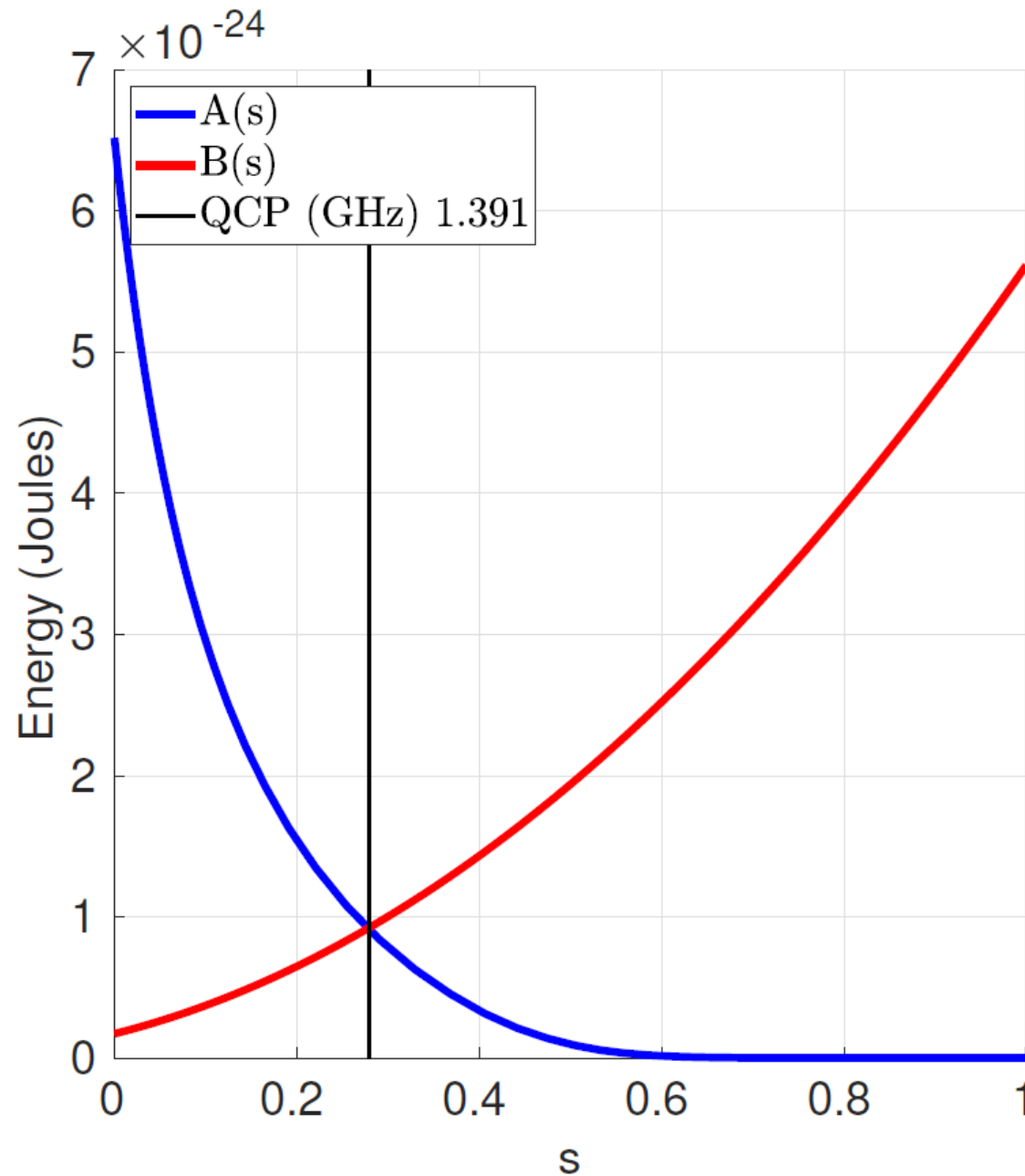
After coding in QA, each σ will become a 0 or a 1, and the optimal solution is where I have friends with high coefficients and enemies with negative coefficients.

SNUBIT



- Ok, how does a DWave processor work?





$A(s)$ = initial configuration. Each qubit is superimposed, some default setting of biases. No couplings needed!

$B(s)$ = problem configuration. Qubits literally quantum tunnel from $A(s)$ to $B(s)$ given a slow change (adiabatic theorem).

S – think of it like time, normalized to go from 0 to 1.

PHEW! Lots of math.

Let's just overview **QA** again, now that we know how it works.



Universal QC (IBM, google, ...)

- Use quantum gates, can model ANY classical algorithm
- Largest processors ~ 120 qubits

DWave's QA

- Just uses a coupled graph for binary optimization. Cannot model any classical algorithm
- Largest processors ~ 5000 qubits



The background features several thin, white, abstract geometric shapes. A large, rounded rectangle is tilted diagonally across the center. In the bottom right corner, there are several nested, stylized arrow-like shapes pointing towards the right.

THAT IS IT

YOU ARE NOW QC
SCIENTISTS, RIGHT?